```
1   FUNCTION_BLOCK FLIP_FLOP
2   VAR_INPUT
3       IN : BOOL; (* Input signal *)
4   END_VAR
5   VAR_OUTPUT
6       Q : BOOL; (* Output signal *)
7   END_VAR
8   VAR
9       fbTrig : R_TRIG; (* Rising edge trigger function block *)
10      xFlip  : BOOL; (* Internal flip-flop state *)
11  END_VAR
12  VAR_TEMP
13      xFlop : BOOL; (* Temporary variable for edge detection *)
14  END_VAR
15
```

```
1   fbTrig(CLK := IN, Q => xFlop);
2
3   (* Temporarily disable SA0062 warning. The static analyser incorrectly
4    flags the R_TRIG function block output as always FALSE. It does not
5    have access to the actual run-time behaviour. *)
6   {analysis -62}
7   IF xFlop {analysis +62} THEN
8       xFlip := NOT xFlip;
9   END_IF
10
11  (* Use a temporary variable for edge detection in order to avoid reading
12   the previous state from the output. *)
13  Q := xFlip;
14
```